

# Aleksei Anisimov

HPC, RUST, SOFTWARE ENGINEER



## Contact

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## Skills

### Programming Languages

Rust, Scala, WGSL, SQL

### Databases, storage formats and a like.

PostgreSQL, Redis, Apache Kafka, Apache Arrow

### Others:

Apache Spark, Slurm, MPI, HPE Cray EX (OLCF), Apptainer / Singularity

## Languages

|         |     |
|---------|-----|
| English | B2+ |
| Serbian | B1  |
| Russian | C2  |

## ABOUT ME

I am a seasoned professional programmer with more than 20 years of experience in various fields and domains.

In recent years, I have worked as a Rust HPC programmer in the biotech field, specializing in optimizing the performance and resource usage of biostatistical and biocomputational pipelines to optimize their runtime characteristics for both Beowulf and supercomputer environments (OLCF). My extensive experience as a Scala developer has given me a solid understanding of functional and type-level programming, which enables me to deliver robust, high-quality software, that builds on the strengths Rust already offers.

Additionally, I have completed graduate-level courses in Bioinformatics, Whole Genome (WGS) and Single-Cell Sequencing, and Genomic Networks. This formal training demonstrates my passion and dedication to delivering top-quality software, as truly effective solutions require a deep understanding of the field.

My efforts, to a sensible extent, helped biologists to develop at least two molecules for treating orphan diseases, and one of them is out of clinical trials.

Besides that I have contributed to the back-end development of a well-known MMO game, an IoT framework, a London minicab automation system, and a bus station automation project.

## SUMMARY

- 10+ years of professional experience in the Information Technology industry;
- Rich experience in Rust, Scala, Python programming;
- Solid background in functional and type-level programming paradigm;
- Development and expertise highlights:
  - HPC application development for supercomputers;
  - Client-server applications development;
  - Continuous Integration management;
  - Advanced technical troubleshooting, application profiling and performance tuning;
- Experience in team leadership.

## EXPERIENCE

### 2019 - Present HPC, Rust programmer BPGBio

The project aim was to learn the structure of biological networks from omics data assessed with patient data using a combined constraint-based approach and Bayesian score-based method (a framework based on S. Boettcher deal package) in a distributed environment.

### Tech. responsibilities

- Prepare SIF containers to run biocomputational pipelines on OLCF supercomputer.
- Develop WGSL compute shaders to run some parts of the computational pipelines on GPGPU in a Beowulf cluster.
- Made an extension for `rsmpi` to support return error codes. And based on those changes I implemented User-Level Fault Mitigation (MPI ULFM) logic to address partial cluster failures.
- Performance optimization and tuning using `vtune`, `mpip`, `heaptrack`, `dhat`, `llvm sanitizers`, `criterion` and `iai` (`cachegrind`);
- Algorithms optimization;
- Porting of R, Python and Wolfram Mathematica scripts, produced by mathematicians to Rust;
- Contributed missing functionality to `special crate` (`tri-gamma` function), `rsmpi` (group splitting by colors and tags, User Level Failure Mitigation);
- Design and implementation of auxiliary programs requested by the analytics department;
- Automation of development and deployment processes (`gitlab-ci`, `podman`);
- HTTP API development using `axum`, `utoipa` and `http4s` for Scala-based services;

- Implementing automated tests (property, unit, mutation, integration and e2e) and benchmarks (criterion, iai);
- Bug fixing in HPC cluster environment (pmpi, vtune);
- Developing and maintaining open source components required by our software: `kendalls` crate for Kendall's tau-b rank correlation, `hoconlight` crate - a peg parser and deserializer of Akka `hocon` configuration files.

#### Non technical

- Onboarding of programmers and scientists;
- Technical-leading and owning the product;
- Helping team members to develop professional development plans;
- Conducting technical interviews;
- Mentoring and conduction summer schools for rust programmers;
- Writing user and programmer guides, software design documents (tectonic, latex, confluence, typst), SOWs.

Stack: Rust, Apache Arrow, BLAS/LAPACK (ndarray-linalg), MPI, Slurm, Apache Kafka

#### 2016 - 2019 **Scala, Rust programmer**

BERG BIOSYSTEMS

Design and Developed Mesos Framework to schedule and execute bioinformatics pipelines. The framework mostly based on Lightbend stack with addition of TypeLevel stack (circe, cats-effect, fs2) for more lightweight utilities and more granular control of task termination.

#### Responsibilities:

- Developing utilities used in biostatistical pipelines;
- Writing user guides, SOWs, RFCs;
- Reviewing DOI:10.1089/cmb.2019.0210, DOI:10.3390/app11062466;
- Coordinating two teams of developers, resolving conflict situations;

Stack: Apache Mesos, Scala 2, Lightbend stack, TypeLevel stack, Apache Kafka

#### 2016-2016 **Senior Developer**

DATAART

The goal of the project was to develop the sales system to be used by call center agents to book hotels for the clients.

Based on different criteria, the system searched hotels in multiple sources providing an agent with all relevant information about a client. The system utilized microservice architecture, which allowed decoupling all logical subsystem, managing and developing them independently of each other.

#### Responsibilities:

- Call dispatching service development that notifies agents about incoming phone calls;
- Integration with internal CRM system;
- Next Caller and Piwik integration;
- Payment gateway integration;
- Back-end RESTful API development.

Stack: Python, PostgreSQL, Tornado, Redis, Django Rest framework, Ansible

#### 2014-2016 **Senior Software Developer**

WARGAMING, GLOBAL MAP

The application allowed clan leaders to plan their strategy on capturing and controlling areas on the global world map, managing clan resources, participating in tournaments and landing battles. The development was focused on back-end optimization.

- Preparation of new game events;
- Fixing bugs;
- Implemented "mini-quests" and corresponding reward, medal system.

Stack: Python, Django, PostgreSQL, Celery, RabbitMQ

#### 2012-2014 **Senior Software Developer**

DATAART

I worked on a web-based dashboard and computational engine that used statistical methods to predict prospective markets for Drug sales.

Stack: ASP.NET, C#, VBA, MS Server

#### 2010-2011 **Software Developer**

DATAART

The application allowed clan leaders to plan their strategy on capturing and controlling areas on the global world map, managing clan resources, participating in tournaments and landing battles. The development was focused on back-end optimization.

- Release management;
- CI/CD pipelines;
- Design and Develop software that integrates ophthalmologic perimeters with EMR system.

Stack: C#, WPF, Silverlight

#### 2007 - 2010 **Software Developer**

KANAR SOFTWARE

I designed and developed the first version of eSigTek - a web-based software to sign documents electronically. One of the features that stood out in the software was the online PDF forms editor. I was responsible for features planning and leading a team of two programmers and a QA.

New feature development and bug fixing for Medical Staff Management and Recruitment System for the USA market. I modernized the front-end part of the system such that it works not only on IE6 and introduced PrototypeJS.

Stack: C#, MS Server, jQuery

#### 2005 - 2006 **Software Developer**

OLVEN-AUTO

I participated in the development of the Bus Station Control System. The developed solution consists of a back-end subsystem that manages bus routes, creates schedules, and exchanges tickets between different bus stations; a front-end that provides GUI to bus station's staff; and a website that customers can use to buy ticks online.

- I worked on the tickets exchange back-end service;
- porting Delphi GUI application to gtkmm;
- integration with a cash register machine, and custom PAM development.

Stack: Firebird, PHP, C++ (gtkmm)

#### 2003 - 2006 **Web Developer**

ZEBRA-GROUP

- Development of modules for the company's internal CMS.

Stack: Python, PHP

## OPEN SOURCE CONTRIBUTIONS

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<https://crates.io/crates/special> — provides special functions for Rust

- Added an implementation of the `tri-gamma` function.

<https://crates.io/crates/mpi> — MPI bindings for Rust

- Added split by node / color functionality.
- Ongoing effort on merging optional ULFM support and errors in function call results.

<https://crates.io/crates/kendalls> — Kendall's  $\tau$ -B

- Maintainer.
- Implemented Kendall's  $\tau$ -b algorithm.

<https://github.com/zolkko/hocon> — lightweight HOCON parser

- Maintainer, developed a parser to ease transition from Lightbend and Type-level stack (Scala) to Rust (`serde_yaml`).

<https://github.com/blumenplace/skju> — eather quake detection BLE mesh network

- Architectural design of the project, and mentoring.
- defmt interceptor for Renode simulator.

- 2016 <https://github.com/ARMmbed/mbed-os>  
• Bug fixes in Arm Mbed OS USB (CDC) for STM32.

## EDUCATION

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- 2010 **Master of Science in Software Engineering**  
VORONEZH STATE TECHNICAL UNIVERSITY  VORONEZH, RUSSIA  
•
- 2007 **Bachelor's Degree in Electrical Engineering**  
ULYANOVSK STATE TECHNICAL UNIVERSITY  ULYANOVSK, RUSSIA  
•
- 2004 **College Degree in Software Engineering**  
ULJANOVSKIY AVIACIONIJ KOLEDZH  ULYANOVSK, RUSSIA

## SIGNIFICANT CERTIFICATES AND AWARDS

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- 2026 **(Ongoing) Machine learning and neural networks in biology and medicine**  
NOVOSIBIRSK STATE UNIVERSITY (NSU)
- 2025 **Genomic networks**  
NOVOSIBIRSK STATE UNIVERSITY (NSU)
- 2022 **From science to drug development**  
BLASTIM
- 2022 **NGS data analysis**  
BLASTIM
- 2021 **Bioinformatics Contest**
- 2019 **Molecular biology**
- 2019 **Third Generation Sequencing using Oxford Nanopore**
- 2012-2016 **Co-organizer of Voronezh's radio hobbyist club**